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RESEARCH ARTICLE

Indacenodithiophene-based single-component ambipolar polymer for high-performance vertical organic electrochemical transistors and inverters

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Abstract

Single-component ambipolar polymers are highly desirable for organic electrochemical transistors (OECTs) and integration into complementary logic circuits with reduced process complexity. However, they often suffer from imbalanced p-type and n-type characteristics and/or stability issues. Herein, a novel single-component ambipolar polymer, namely, gIDT–BBT is reported based on indacenodithiophene (IDT) as the electron donor, benzobisthiadiazole (BBT) as the electron acceptor and oligo ethylene glycol (OEG) as the side chain. Benefitting from the extended backbone planarity and rigidity of IDT, pronounced electron-withdrawing capability of BBT, favored ionic transport from OEG together with vertical OECT device structure, a nearly balanced ambipolar OECT performance is achieved for gIDT–BBT, revealing a high transconductance of 155.05 ± 1.58/27.28 ± 0.92 mS, a high current on/off ratio $>10^6$ and an excellent operational stability under both p-type and n-type operation conditions. With gIDT–BBT in hand, furthermore, vertically stacked complementary inverters are successfully fabricated to show a maximum voltage gain of 28 V V^{-1} (V_{IN} = 0.9 V) and stable operation over 1000 switching cycles, and then used for efficient electrooculogram recording. This work provides a new approach for the development of ambipolar single-component organic mixed ionic–electronic conductors and establishes a foundation for the manufacture of high-performance ambipolar OECTs and associated complementary circuits.

KEYWORDS
electrooculogram recording, inverters, single-component ambipolar polymer, vertical organic electrochemical transistors

1 | INTRODUCTION

Organic electrochemical transistors (OECTs) can convert ionic-related biological signals into electronic signals via electrochemical doping of the active polymer channel by the electrolyte ions, and thus hold a great potential for the growing bioelectronics, such as chemical/biological sensing[1–4] electrophysiological recording[5,6] and neuromorphic computing.[7–9] To this end, lots of organic mixed ionic–electronic conductors (OMIECs) with a conjugated

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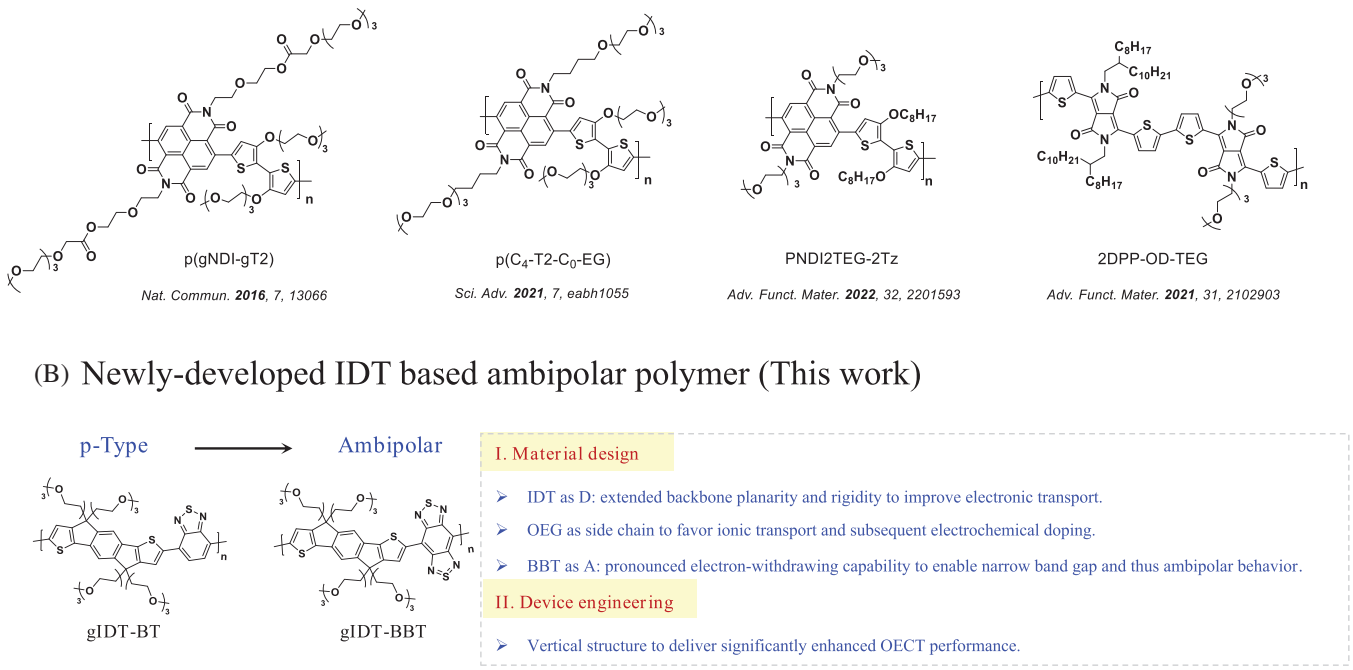


FIGURE 1 Chemical structures of (A) reported ambipolar organic mixed ionic–electronic conductors (OMIECs) and (B) gIDT–BT and gIDT–BBT.[15–18] BBT, benzobisthiadiazole; DPP, diketopyrrolopyrrole; IDT, indacenodithiophene; NDI, naphthalene diimide; OECT, organic electrochemical transistor; OEG, oligo ethylene glycol.

main chain and a hydrophilic oligo ethylene glycol (OEG) side chain have been successfully designed toward conducting ionic and electronic transport simultaneously, and used for efficient p-type, n-type, and ambipolar OECTs.[10–13] Among them, ambipolar OECT materials capable of transporting both holes and electrons are highly desirable, because they could be integrated into the complementary logic circuits with improved signal detection sensitivity, facilitated implementation of multifunction, and reduced process complexity.[12]

One strategy to develop ambipolar OECT materials is the utilization of bulk-heterojunction blends.[14] In this case, p-type and n-type polymers are mixed together in a single layer to generate a bicontinuous network, enabling the transport of different charge carriers. Although a balanced ambipolar characteristic is realized, such a blending technology inevitably suffers from the precise control of phase separation and reproducibility issues. As an alternative, single-component ambipolar polymers are proposed by combining an electron donor (D) and an electron acceptor (A) to well tune the band gaps.[12] However, only a few ambipolar OMIECs have been reported. And the majority of previously reported single-component ambipolar polymers are based on the naphthalene diimide (NDI) building block (Figure 1A).[15–17] Owing to the twisted polymeric backbone, all of them behave either imbalanced or low ambipolar properties. Besides, a diketopyrrolopyrrole (DPP)-based polymer 2DPP–OD–TEG[18] with a good backbone planarity is reported to exhibit a similar current on/off ratio (I_{ON}/I_{OFF}) of 10^5 and a comparable transconductance (g_m) of 0.084 mS and 0.196 mS for n-type and p-type modes, respectively. However, its p-type stability (current dropped by 90% under 80 cycles) seems to be much poorer than that of the n-type mode (current unchanged under 80 cycles). Altogether the imbalanced properties between p-type and n-type modes, obtained for NDI- and DPP-based ambipolar polymers, severely limit their practical application in the complementary OECT logic circuit integration. Therefore, much effort should be paid into the development of single-component ambipolar polymers, which show high and matched p-type and n-type OECT performance including transconductance, current on/off ratio, and stability.

With this idea in mind, we newly develop an indacenodithiophene (IDT)-based single-component ambipolar polymer gIDT–BBT for high-performance vertical OECTs and inverters. As depicted in Figure 1B, IDT containing a five fused-ring structure is selected as the D. Compared to NDI and DPP building blocks, the extended backbone planarity and rigidity is anticipated to improve electronic transport.[19,20] Second, OEG is introduced into IDT as the side chain in order to favor ionic transport and subsequent electrochemical doping. Third, benzobisthiadiazole (BBT) instead of benzothiadiazole (BT) serves as the A to form a D-(strong)A geometry.[21,22] The intrinsic pronounced electron-withdrawing capability with a distinctive double-free-radical nature is beneficial for the generation of quinone-type resonance structure along the polymeric backbone and strong intramolecular charge transfer (ICT), endowing gIDT–BBT with a narrow band gap of 0.85 eV and thus a typical ambipolar behavior. To further improve and balance p- and n-type device performance, a vertical OECT (vOECT) structure was applied, which could offer higher transconductance (especially for n-type performance), decreased transistor footprint, and increased recording density, simultaneously, in comparison with that of planar OECT structure.[23–29]

As a result, gIDT–BBT-based vOECT achieves a nearly balanced p-type and n-type OECT performance, revealing

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TABLE 1 Summary of the organic electrochemical transistor (OECT) performances based on ambipolar materials.[15–18]

Materials		I_{ON} (mA)	I_{ON}/I_{OFF}	g_m (mS)	Cyclings and remained I_D compared to initial I_D	Channel width/length (μm)	Reference
p(gNDI-gT2)	n	3.80×10^{-3}	32000	0.021	360, 100%	100/10	Nat. Commun. 2016, 7, 13066
	p	-1.40×10^{-3}	2010	0.013	–		
p(C4-T2-C0-EG)	n	–	–	1.53×10^{-4}	–	100/10	Sci. Adv. 2021, 7, eabh1055
	p	–	–	1.36×10^{-4}	–		
PNDI2TEG-2Tz	n	11	1×10^5	51.8	–	1800/5	Adv. Funct. Mater. 2022, 32, 2201593
	p	–0.5	1×10^4	–	–		
2DPP-OD-TEG	n	0.01	1×10^5	0.084	80, 100%	100/10	Adv. Funct. Mater. 2021, 31, 2102903
	p	–0.02	1×10^5	0.196	80, 10%		
gIDT-BBT	n	4.19	1×10^6	27.28	1000, 85%	30/30	This work
	p	–38.95	1×10^6	155.05	2000, 91%		

a high g_m of 155.05 ± 1.58/27.28 ± 0.92 mS, a high current on/off ratio $>10^6$ with a small channel area footprint of 30 × 30 μm under both p- and n-type operation conditions. In addition, excellent operational stabilities under both positive and negative bias for 2000 cycles are obtained for gIDT–BBT-based vOECT. These values outperform previous single-component ambipolar polymers based on NDI or DPP (Table 1). Furthermore, its corresponding vertically stacked complementary inverters (VSCIs) are also demonstrated to give a maximum voltage gain of 28 V V^{-1} (V_{IN} = 0.9 V) and stable operation over 1000 switching cycles, which applied used as a voltage amplifier of electrooculography (EOG) signal successfully.

2 | RESULTS AND DISCUSSION

To gain a comprehensive understanding of the geometric and electronic structures of the synthesized materials, density functional theory (DFT) calculation based on Gaussian 09 method (B3LYP/6-31G) was conducted.[30,31] The ethylene glycol side chains are regarded as methyl groups to simplify the calculation process. The optimized geometries of the two trimeric structures and their frontier molecular orbitals are depicted in Figure 2A. Both gIDT–BT and gIDT–BBT trimers have virtually planar conformation, with dihedral angles of approximately 179.92° and 179.99° between the IDT unit and the neighboring acceptors, respectively. Such rigid and coplanar structure is beneficial for charge transfer within the polymer backbone and improving electron mobility. The visualization of lowest occupied molecular orbitals (LUMOs) and highest occupied molecular orbitals (HOMOs) is illustrated in Figure 2B. Both trimers exhibit localized electronic structures, with donor units mostly contributing to HOMOs and LUMOs predominantly localized on the electron-deficient BT or BBT units. The E_{LUMO}/E_{HOMO} values obtained from DFT simulation are −2.69/−4.50 eV for IDT-BT and −3.53/−4.33 eV for gIDT–BBT, respectively. The deeper LUMO and shallower HOMO in gIDT–BBT, as well as the corresponding narrower bandgap, originated from the stronger electron-withdrawing ability of the BBT unit and quinone-type resonance characteristics, which is consistent with previous literature report.[21,22]

The detailed synthesis processes of both monomers and copolymers can be found in Supporting Information (Scheme S1). Note, by integrating four triethylene glycol side chains, excellent ion conductivity in aqueous environments and good solubility in organic solvents (including chloroform and dichloromethane) are obtained in the IDT-based copolymers.[11,16,25] Moreover, the resulting polymers are purified by consecutive Soxhlet extractions using methanol, *n*-hexane, and acetone to remove low molecular weight components and contaminants. The purified polymers were extracted with chloroform, precipitated with methanol, and further vacuum dried. The Nuclear magnetic resonance (NMR) for ^1H and ^{13}C was used to characterize chemical structures (Figures S1–S6). The number-average molecular weights (M_n, 11.9 kDa and 7.7 kDa) and polydispersity index (3.61 and 1.49) for gIDT–BT and gIDT–BBT, respectively, were determined by gel permeation chromatography with trichloromethane as the eluent and polystyrene as standard (Figure S7).

Next, ultraviolet–visible–near-infrared (UV–vis–NIR) absorption spectra of both polymers in solution (chloroform as solvent) or in film state are characterized (Figures S9 and 2C). Both polymers show typical dual-band absorption characteristics of the D–A skeleton, including ICT absorption in the low-energy region (600–1200 nm) and π–π^* transitions in the high-energy region (300–500 nm). Notably, gIDT–BBT film exhibited a distinct redshift absorption peaks and broader ICT band around 1045 nm, which is attributed to increased electron-withdrawing ability and BBT quinone-type resonance. Optical bandgaps ($E_{g\ opt}$) calculated from film absorption onsets are 1.68 and 0.85 eV for gIDT–BT and gIDT–BBT, respectively. In addition, electron paramagnetic resonance (EPR) spectroscopies of gIDT–BT and gIDT–BBT solids are also measured at room temperature. A sharp EPR signal at g = 2.004 is detected for gIDT–BBT, whereas gIDT–BT is EPR silent (Figures 2D and S8), indicating a high spin ground state or quinone-type resonance in gIDT–BBT. Therefore, the comparatively stronger ICT feature

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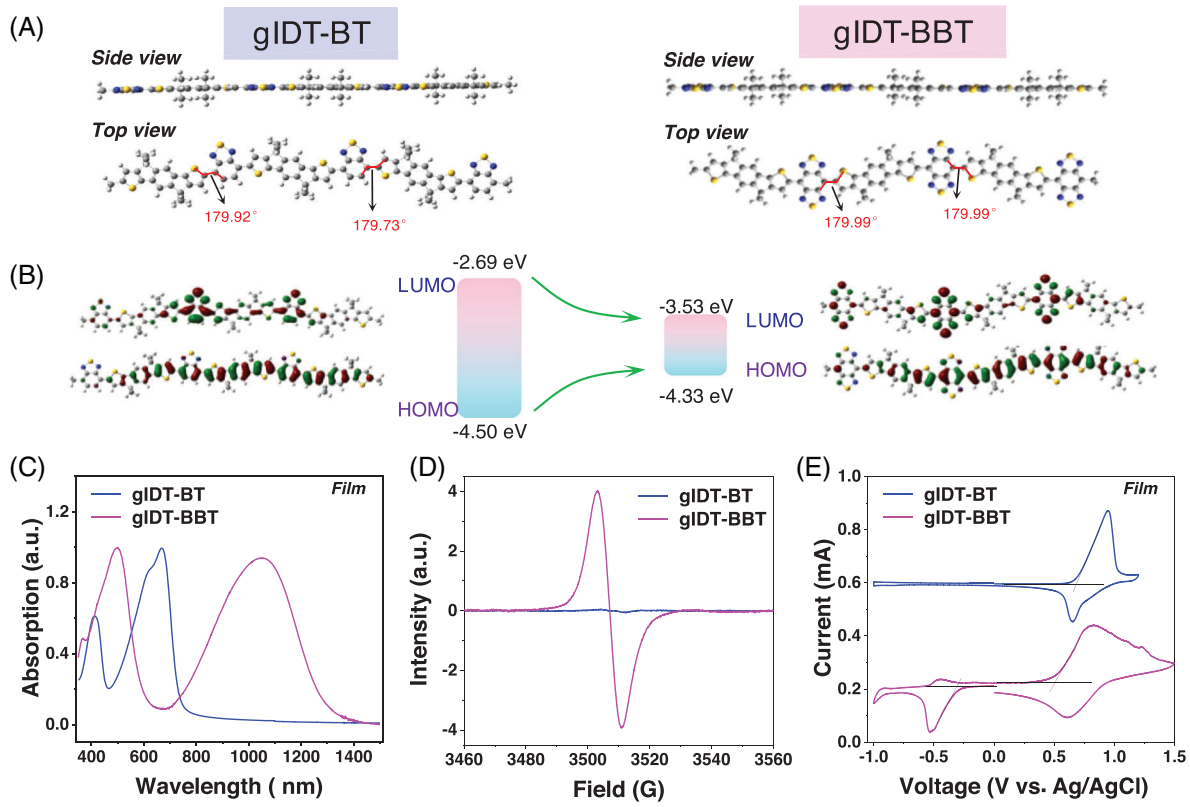


FIGURE 2 (A) Trimer structures of gIDT–BT and gIDT–BBT calculated by density functional theory (DFT). (B) Frontier molecular orbitals and calculated highest occupied molecular orbital (HOMO)/lowest occupied molecular orbital (LUMO) energy levels. (C) Ultraviolet (UV) absorption spectra of the polymer films. (D) Room temperature electron paramagnetic resonance (EPR) characteristics of the polymers. (E) Cyclic voltammetry of the polymer films in acetonitrile electrolyte.

and quinone resonance are responsible for the significantly narrower bandgap of gIDT–BBT.

Electrochemical properties and energy levels of both polymers are further studied through cyclic voltammetry (CV) characterization, where $CHCN_3$ containing 0.1 M tetrabutyl-ammonium hexafluorophosphate (Bu_4NPF_6) acts as electrolyte and platinum plate as working electrode. As shown in Figure 2E, gIDT–BT only has a quasi-reversible oxidation peak, while gIDT–BBT exhibits distinct reduction and oxidation peaks. Consequently, it is indicated that gIDT–BBT is more oriented toward ambipolar characteristics, while gIDT–BT is only inclined toward p-type. Moreover, E_{HOMO}/E_{LUMO} estimated from CV measurements is −5.01/−3.33 eV for gIDT–BT and −4.80/−4.03 eV for gIDT–BBT (Table S1), which agrees well with the DFT calculation and UV–vis–NIR characteristic.

An ideal OMIEC should be electrochemically stable in aqueous environments.[10,32] Considering the stability of doped states of OECT materials, their HOMO and LUMO levels should be less negative than −5.25 eV and more negative than −4.02 eV (gray-shaded area with a black dashed line in Figure 3A), corresponding to the potential for oxygen evolution and water reduction (at pH = 7), respectively.[15,18] For gIDT–BT, the LUMO level (−3.33 eV) is outside of the electrochemical stable window. While for gIDT–BBT polymer, both HOMO (−4.80 eV) and LUMO (−4.03 eV) levels are within the electrochemically stable window, which is expected to show stable ambipolar OECT operation in aqueous electrolyte. The electrochemical doping processes and stability of gIDT–BT and gIDT–BBT in aqueous media are further investigated by CV measurement with phosphate buffer solution [PBS (1×)] as electrolyte (Figures 3B,C and S10). The gIDT–BT only exhibited an oxidation peak corresponding to the p-doping state, while gIDT–BBT demonstrated both oxidation and reduction peaks that correspond to p- and n-doping. What is more, both gIDT–BT and gIDT–BBT exhibited excellent stability, which show 80 cycles of reversible doping and dedoping processes in air without the removal of oxygen.

UV–vis–NIR spectroelectrochemical measurements in PBS aqueous electrolytes were also conducted to study the ion doping of both polymers. As illustrated in Figures 3D–G and S11, spectral changes have been measured by applying different voltages (detailed experimental setup and process can be found in Section 4). When a gradually increasing positive voltage (0 to 0.8 V, Figure 3D) is applied at Ag/AgCl electrode, almost identical absorption spectra of gIDT–BT are observed, indicating inefficient n-type doping. When a negative voltage (0 to −1.3 V, Figure 3E) is applied, the intensity of the ICT absorption peak (~670 nm) diminishes slightly along with obvious strengthen of the polarization peak at 1050 nm. These results suggest that gIDT–BT is far more likely to be p-doped. In contrast, for gIDT–BBT, when positive voltage (0 to 0.9 V, Figure 3F) and negative voltage (0 to −0.7 V, Figure 3G) are applied, it exhibits efficient p-doping and n-doping, as evidenced by the significantly diminished neutral absorption features and increased polaron absorption around 1600 nm (Figure S11).

OECT performances are first investigated with conventional planar structure, where channel width (W) and length (L) are 500 and 70 μm, respectively (Figure S12, detailed fabrication process can be found in Section 4).

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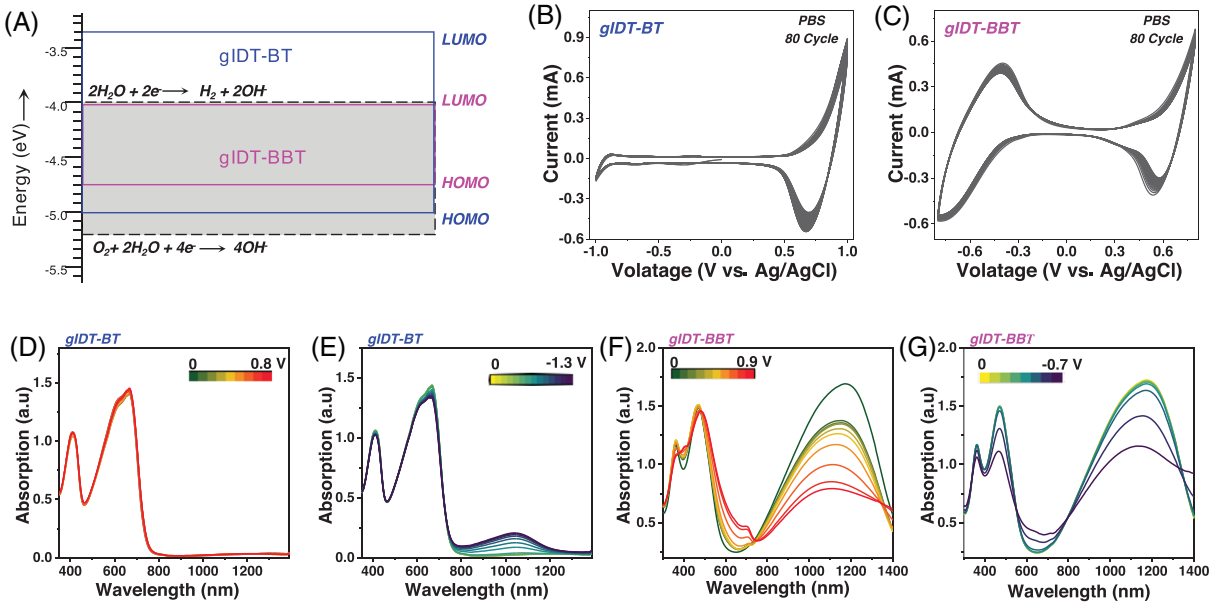


FIGURE 3 (A) Redox reaction window of polymers in PBS aqueous solution compared with the electrochemical stabile window (gray area). Cyclic voltammetry of gIDT–BT (B) and gIDT–BBT (C) with a scan rate of 100 mV s^{-1} in PBS (1×) aqueous solution (80 cycles). Spectroelectrochemical characteristics of (D, E) gIDT–BT and (F, G) gIDT–BBT films under positive or negative bias in PBS aqueous solution.

Representative transfer and output curves of the OECTs are shown in Figures S13 and S14, with corresponding OECT parameters summarized in Table S2. It is obvious that gIDT–BT only exhibits p-type (a maximum drain current [I_{ON}, I_D under $V_G = -0.68$ V] of 130.48 ± 2.51 μA), but not n-type characteristics. In contrast, gIDT–BBT-based OECTs demonstrate pronounced ambipolar OECT behavior, with a p-type maximum g_m of 0.78 ± 6.2E^{-2} mS (I_{ON} of 156.45 ± 1.69 μA), and a weaker n-type maximum g_m of 0.02 ± 9.6E^{-4} mS (I_{ON} of 5.04 ± 0.15 μA). These results are comparable to previous literature reports.[15–18]

Next, vertical OECT structure is introduced to further exam the performances of both polymers, since vOECT has been proved to deliver significantly enhanced performance, especially for n-type ones, due to enhanced electron injection.[24] Fabrication process of vOECT is illustrated in Figure S15, and details can be found in Section 4. Note, here, a photocurable agent, namely, DtFDA (a redox-inactive small molecular, Figure S16),[33,34] is blended with the OMIEC for the following reasons: (1) Patterning, which allows for well-defined channel dimensions; (2) Higher stability, which is accomplished through chemically cross-linked networks that prevent channel swelling and delamination of top electrode after repeated ionic doping and electrolyte injection/ejection. Moreover, the incorporation of a redox-inactive photocurable crosslinker may also facilitate charge transfer and ion transport in the OMIEC:crosslinker blend.

Therefore, the weight ratios of OMIEC and DtFDA are first finetuned from 8:1 to 7:1, 6:1, 5:1, and 4:1. Film morphologies and microstructures of the blends are first characterized by atomic force microscopy (AFM) and two-dimensional grazing-incidence wide-angle x-ray scattering (2D-GIWAXS). AFM images (Figures S17 and S18) reveal that gIDT–BBT, gIDT–BT, and DtFDA neat films hold a smooth and continuous surface with root mean square roughness (σ_{RMS}) of 0.56, 0.24, and 0.69 nm, respectively. With the increasing content of DtFDA in gIDT–BBT, σ_{RMS} gradually increases from 0.65 to 0.78. 2D-GIWAXS patterns reveal that both neat polymers and blends have amorphous morphology (Figures S19 and S20), which is consistent with IDT-based polymers.[35] Consequently, it is concluded that the amount of DtFDA in the blend show minor effects on film morphology, crystallinity, and packing features.

Next, vOECT performances based on various OMIEC:DtFDA blends are characterized (Figures S21–S30) and summarized in Table S3 and Figure S31. All gIDT–BT-based vOECTs only exhibit p-type performance with I_{ON} of 2.05 ± 0.15 mA (at $V_{DS} = -0.2$ V, $V_G = -0.7$ V) and maximum g_m of 27.44 ± 0.42 mS. In contrast, gIDT–BBT-based vOECTs demonstrate apparent ambipolar characteristics, where I_{ON} of −32.54 ± 2.40 mA (at $V_{DS} = -0.2$ V, $V_G = -0.7$ V) and g_m of 170.42 ± 6.06 mS are obtained under p-type mode, along with I_{ON} of 8.22 ± 1.24 mA (at $V_{DS} = 0.2$ V, $V_G = 0.7$ V) and g_m of 33.81 ± 2.51 mS under n-type mode. It is worth noting that vOECTs exhibited 1000 times higher I_{ON}, ~200 times higher g_m, and significantly larger I_{ON}/I_{OFF} ratios (≈10^6) than these in corresponding planar OECTs. Such performance (g_m > 33 mS with I_{ON}/I_{OFF} ratio >10^5) represents the highest reported values among single-component ambipolar OECTs to date. It is also observed that, as the DtFDA content increases, both I_{ON} and g_m for n-type performance decrease gradually (Figure S31). While the hysteresis between the forward and the reverse sweeps significantly decreases and becomes negligible when DtFDA content is higher than 12.5% (gIDT–BBT:DtFDA < 7:1; Figures S21 and S23), indicating enhanced ionic transport after DtFDA introduction. Compared to planar structures, the performance of vertical OECTs has shown significant enhancement and optimization. For vOECTs, the semiconductor material is sandwiched between the two electrodes (Figure 4A). First, the contact resistance between the organic semiconductor (OSC) and the electrodes is reduced, thereby enhancing the injection of electrons and holes.[36] Second, the contact area between oxygen/water and the semiconductor is decreased, resulting in less electron traps. Most importantly, compared to planar transistors

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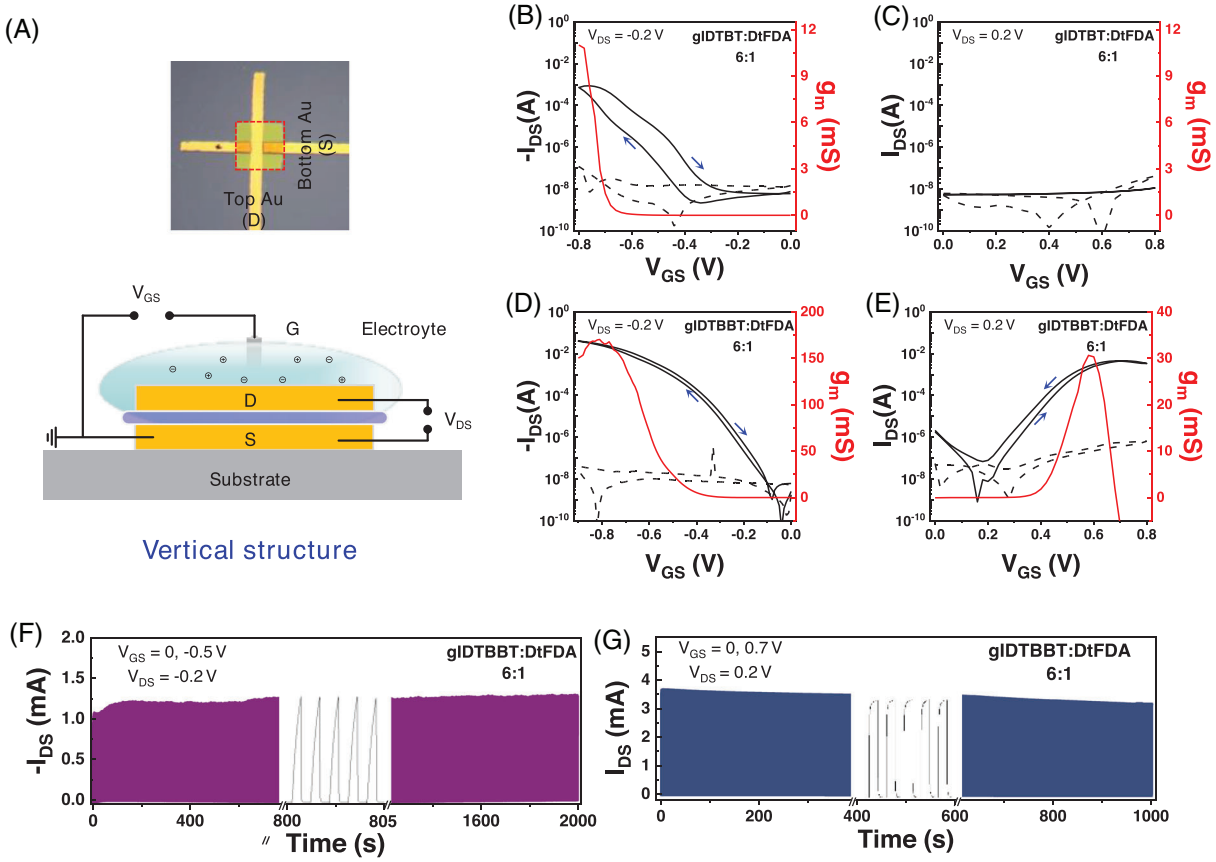


FIGURE 4 (A) Schematic diagram of vertical organic electrochemical transistor (OECT) device structure (bottle) and optical microscope image (top). (B, C) Transfer and g_m curves of vOECTs-based gIDT–BT:DtFDA (6:1) and (D, E) gIDT–BBT:DtFDA (6:1) blend films under (B, D) p-type mode and (C, E) n-type mode. The dash lines present gate current, and the g_m curves are calculated from the forward sweeps. Cycling stability of gIDT–BT:DtFDA (6:1) based vOECTs under (F) p-type and (G) n-type mode with a pulse width of 0.5 and 0.25 s, respectively.

with channel length around 10 μm, by employing a vertical structure the channel length "L" is significantly shortened to nanoscale. Therefore, the output performance is improved by 2–3 magnitude orders, according to the formula ($g_m = (Wd/L)\mu C^*(V_G - V_{Th})$).[11]

Transient characteristics of the ambipolar vOECTs are carried out and depicted in Figures S26 and S27. As the DtFDA content increases, both turn-on transient time (τ_{ON}) and turn-off transient time (τ_{OFF}) gradually decrease (Table S3). For instance, vOECTs based on gIDT–BBT:DtFDA = 6:1 exhibited fast switching behavior with τ_{ON}/τ_{OFF} of 36.44/20.15 ms in n-type mode and 170.33/9.97 ms in p-type mode, which are much faster than these in vOECTs based on neat gIDT–BBT (τ_{ON}/τ_{OFF} of 670/570 ms and 490/120 ms in n-type and p-type mode, respectively). The decrease of τ_{ON}/τ_{OFF} indicates more efficient ion extraction process, and such trend is consistent with the decreasing hysteresis in transfer curves. This phenomenon might be attributed to less content of gIDT–BBT in blends, which facilitates ion injection and extraction processes. Cycling stability characteristics are further performed by applying a square wave of V_G (from 0 to −0.5 V under constant V_{DS} of −0.2 V, p-type mode; or from 0 to 0.7 V, under constant V_{DS} of 0.2 V, n-type mode). As depicted in Figures S28 and S29, neat gIDT–BBT-based vOECTs exhibit poor cycling stability, with complete loss of output current occurring unexpectedly during an uncertain cycle, which is attributable to active layer delamination after swelling. While vOECTs with DtFDA crosslinker show dramatically enhanced stability. For instance, ambipolar vOECTs based on gIDT–BBT:DtFDA = 6:1 (Figure 4B–G) demonstrated consistent performance (maintaining 91.45% and 85.39% of I_{ON} under both p-type and n-type modes, respectively) after 1000 cycles. When compared to previously reported ambipolar OECTs (Figure 5A,B and Table 1), our ambipolar vOECTs based on gIDT–BBT:DtFDA = 6:1 reveal outstanding performance with negligible hysteresis, relatively balanced n-type and p-type g_m, fast transient response, and excellent stability. This offers the groundwork for the development of stable and high-performance ambipolar OECT logic circuits.

Key features for the next generation of biomedical devices are low cost, high amplification capability, and low power consumption.[37,38] OECT-based inverters and their logic circuits possess a remarkable ability for voltage amplification, enabling electrophysiological signals to be detected and amplified, simultaneously. Furthermore, their low operating voltage (<1 V) ensures low power consumption and safety, making them an ideal choice for electrophysiological signal sensing. Herein, complementary inverters are fabricated by vertically stacking two ambipolar vOECTs (Figures 6A and S32). Such three-dimensional architecture enables increased integration density, leading to a 50% reduction in the footprint area of an inverter. Ambipolar semiconductor can behave as both p-type and n-type transistors when the effective V_G is negative and positive, respectively. During inverter measurement (the schematic diagram of the simulated circuit is depicted in Figure S33), a bias of 0.7 V for V_{DD} and

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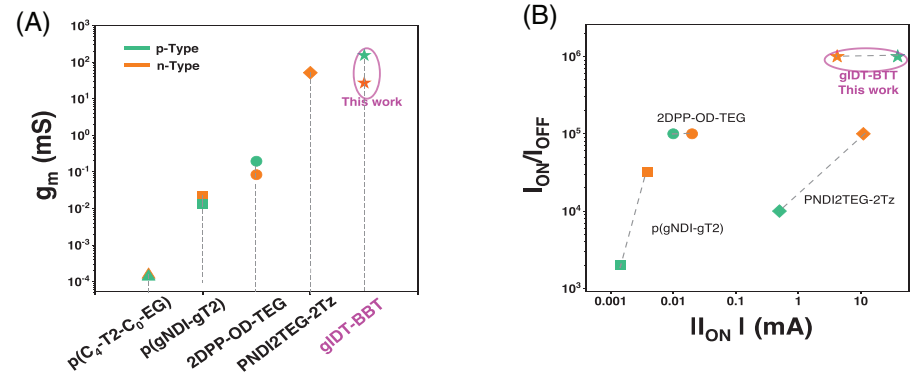


FIGURE 5 Comparisons of the performance for our work as well as the previously reported ambipolar organic electrochemical transistors (OECTs), which about (A) g_m and (B) current on/off ratio (I_{ON}/I_{OFF}) versus I_{ON}.[15–18]

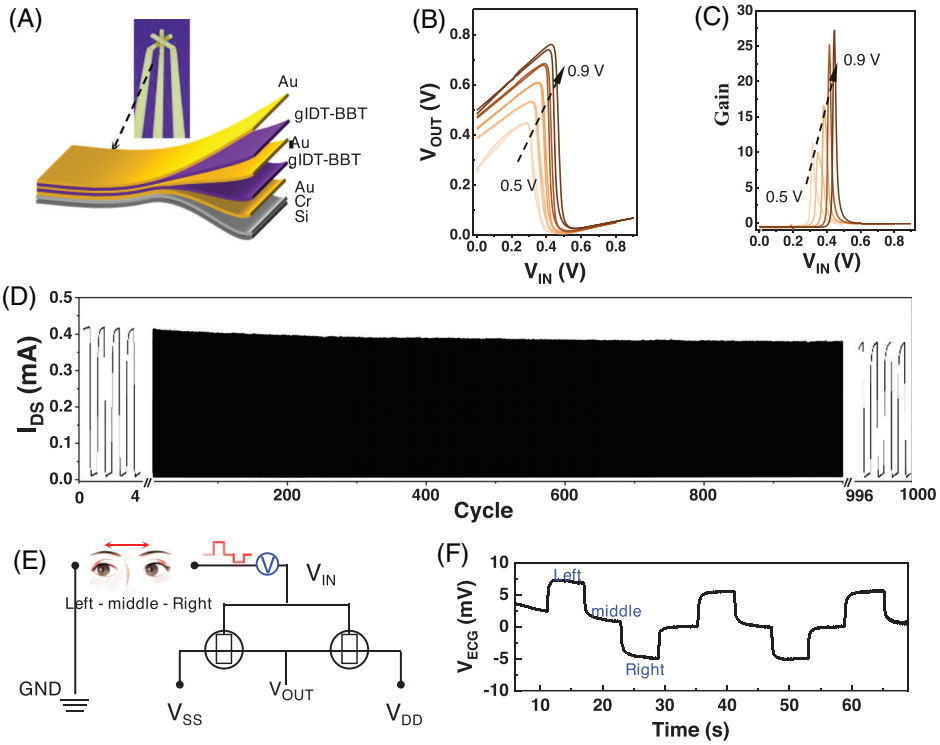


FIGURE 6 (A) Illustration of the ambipolar vertical organic electrochemical transistor (vOECT)-based complementary inverter. (B) Representative voltage transfer curves, (C) the associated voltage gains, and (D) the cycling stability of the complementary inverter. (E) Illustration of the circuit during electrooculography (EOG) signal monitoring, and (F) the amplified EOG signals.

0 V for V_{SS} was applied. The effective V_G for bottom transistor connected to V_{DD} is −0.7 to 0.2 V when V_{IN} from 0 to 0.9 V, resulting in p-type mode from on to off. While the V_G for top transistor connected to V_{SS} is increased from 0 to 0.9 V, resulting in n-type mode from off to on. The voltage transfer and corresponding voltage gain characteristics under different V_{IN} are illustrated in Figure 6B,C, where a sharp voltage transition with a gain of approximately 28 V V^{-1} (under V_{IN} = 0.9 V) is obtained, along with stable switching for over 1000 cycles.

Furthermore, the potential of the inverter for the amplification of EOG signals is investigated. Note, the EOG signals are very weak (with only ∼1.5 mV potential variations induced by horizontal [left-to-right] or vertical [up-to-down] eye movements).[39] Two 3M electrodes are connected to the left and right sides of the eye, while one electrode is linked to the Ag/AgCl gate of the inverter and the other to the ground (Figure 6E). The EOG signals are recorded with V_{DD} of 0.23 V and V_{SS} of −0.47 V. The output voltage fluctuated significantly and consistently over three cycles of left-to-right eye movements (Figure 6F). The inverter recorded a maximum 10 mV variations when the eye moved from left to right. Overall, this ambipolar vertical complementary inverter performed admirably in monitoring eye movements, emphasizing its enormous potential as a signal amplifier in wearable integrated systems.

3 | CONCLUSION

In conclusion, a novel single-component ambipolar polymer gIDT–BBT has been developed for high-performance vertical OECTs and inverters by using a five-ring-fused IDT as the D, a strong electron-withdrawing BBT as the A, and a hydrophilic OEG as the side chain. Through material design and device engineering, matched p-type and n-type

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characteristics are realized for gIDT–BBT-based ambipolar vOECTs, demonstrating a record-high g_m, I_{ON}/I_{OFF}, and operational stability. Moreover, such ambipolar vOECTs are further integrated into efficient VSCIs to show a maximum voltage gain of 28 V V^{-1} (V_{IN} = 0.9 V) and steady operation over a thousand cycles. Remarkably, the obtained VSCIs can be successfully applied to amplify EOG signals, exhibiting an extraordinary eye-tracking performance. This work, we believe, not only establishes an innovative approach for single-component ambipolar polymers, but also presents a paradigm for high-performance ambipolar OECTs and associated complementary logic circuits.

4 | EXPERIMENTAL SECTION

4.1 | Material preparation

The supporting information contains details on the molecular synthesis and characterization. The compounds 4,9-dihydro-s-indaceno[1,2-b:5,6-b′]dithiophene and 4,7-dibromo-benzo[1,2-c:4,5-c′]bis ([1,2,5]thiadiazole) were obtained from the GR-chem platform, while other chemicals such as triethylene glycol monomethyl ether, tosyl chloride, sodium hydroxide, sodium iodide, *n*-butyllithium (2.5 M in hexane), trimethyltin chloride, tris(dibenzylideneacetone)-dipalladium [$Pd_2(dba)_3$], tris(ortho-methylphenyl)phosphine, deuterated reagents, and anhydrous acetonitrile were purchased from Innochem company and utilized as received. According to previous work,[33] the photocurable agent DtFDA was synthesized and provided by Prof. L. W. Feng, and will be published elsewhere. We used PBS buffer quick-dissolving particles purchased from Jiangsu Baishi Bio-Technology Co., Ltd., which were diluted with distilled water according to their configuration requirements to prepare the PBS (1×) solution.

4.2 | Material characterization

The ^{1}H NMR and ^{13}C NMR spectra were acquired on Bruker AV-400 spectrometers at 298 K with tetramethyl silane as reference and reported in parts per million. CV measurements were conducted using a CHI760E A21731 instrument, employing a standard three-electrode configuration. A platinum foil served as the working electrode, while a platinum wire electrode and an Ag/AgCl electrode were used as the counter and reference electrodes, respectively. The measurements were carried out in either a deoxygenated, anhydrous 0.1 M tetra-*n*-butylammonium hexafluorophosphate ($TBAPF_6$) acetonitrile solution or a PBS (1×) aqueous electrolyte. The scan rate was set at 100 mV s^{-1}. The LUMO and HOMO levels are calculated by $E_{LUMO} = -(4.8 - E_{Fc/Fc^+} + E_{red,org})$ and $E_{HOMO} = -(4.8 - E_{Fc/Fc^+} + E_{ox,org})$.[40]

4.3 | Spectroelectrochemistry measurements

The polymer thin films were spin-coated on ITO glass before being immersed into a cuvette which is filled with PBS (1×) solution electrolyte, together with an Ag/AgCl electrode. The UV–vis–NIR spectra of polymer films were monitored in situ at different voltages (vs. Ag/AgCl) to probe the neutral and charged states during electrochemical doping (KEITHLEY 2401 Source Meter). A UV-3600i Shimadzu UV–vis–NIR spectrophotometer was utilized with the light beam passing through the cuvette containing PBS (1×) aqueous solution and the polymer/Indium tin oxide (ITO) glass sample. Before the experiment, a background spectrum was recorded using a cuvette/electrolyte/ITO glass slide for baseline correction. After applying the potential for 5 s, a UV–vis spectrum was obtained.

4.4 | Electron paramagnetic resonance spectroscopy

EPR spectroscopy was conducted on a Bruker EMX PLUS. GIWAXS measurements were obtained at 1W1A Diffuse X-ray Scattering Station, Beijing Synchrotron Radiation Facility (BSRF-1W1A). The wavelength of the monochromatic light source of the light source was 1.54 Å. The data were recorded by using the two-dimensional image plate detector of Eiger 2M from Dectris.

4.5 | Device preparation and characterization

4.5.1 | Planar OECT fabrication and characterization (schematic diagram of the preparation process is shown in S10)[24]

The procedure involves employing a template method to cover a mask template onto the surface of a silicon substrate, followed by utilizing vapor deposition to deposit chromium (Cr) and gold (Au) onto the surface of the silicon substrate at a rate of 0.1 Å s^{-1} to achieve a thickness of 3 nm and 1 Å s^{-1} to achieve a thickness of 60 nm, respectively. After removing the mask template, patterned gold/chromium electrodes are obtained as the source and drain electrodes, with a channel size of W/L = 500 μm/70 μm, where W represents the width of the electrode, L represents the distance between two electrodes, and d refers to the thickness of film. A polymer material is dissolved in chloroform with a concentration of 10 mg/mL and spin-coated on the patterned substrate using a volume of 30 μL with spinning speed of 3000 rpm. To perform testing, Ag/AgCl is utilized as the gate electrode and vertically inserted into a PBS (1×) electrolyte. The electrical characterization of the OECTs and inverters was carried with a PRIMARIUS semiconductor parameter analyzer (model: FS373) in ambient conditions. The voltage sweeping speed was 0.1 V s^{-1} for the OECT measurements.

4.5.2 | Vertical OECT fabrication and characterization (preparation process diagram is shown in S13)[24]

The bottom electrode is fabricated using the same process as the planar construction, consisting of a 3 nm layer of chromium (Cr) and a 60 nm layer of gold (Au). To create the

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desired pattern for the active layer, we combine the semiconductor polymer material with the photosensitive crosslinking agent DtFDA in different weight ratios and dissolve them uniformly in chloroform (10 mg mL^{-1}). Subsequently, the resulting mixture is spin-coated onto the substrate surface that has been previously prepared. By using a mask template and exposing it to UV light with a wavelength of 365 nm for 2 min, we selectively crosslink the active layer. Afterward, the excess material is removed through chloroform cleaning, resulting in a precisely patterned active layer with specific dimensions. The positioning of the top electrode is aligned according to the mask template, and a 60 nm thick layer of gold is deposited as the top electrode through thermal evaporation. Additionally, the overlapping area of the electrodes is increased, with the widths of the bottom and the top electrodes define the channel width (W) and the nominal depth (d) of the semiconductor at 30 μm. To create an encapsulating layer, a mixture comprising a photosensitive crosslinking agent with a weight ratio of 4:1 between the cinnamate–cellulose polymer (Cin–Cell)[24] and DtFDA is applied onto the prefabricated device surface via spin coating, using 30 μL of the mixture spun at 5000 rpm. The patterned photo-crosslinking is executed using a mask template and UV light, while the encapsulation layer undergoes a 2-min exposure to UV light with a 365 nm wavelength to achieve patterned crosslinking. As the standard vertical structured OECT testing devices, Ag/AgCl is vertically inserted into a PBS (1×) aqueous electrolyte solution.

4.6 | Inverter preparation and application

Two stacked vOECTs make up the vertical inverter with an inverted structure. The preparation procedure is identical to that of a vOECT, in that the bottom electrode is patterned via mask deposition. UV light-induced crosslinking and chloroform cleaning are used to create patterned structures in the first layer of the active layer. After depositing another layer of a gold electrode, a second layer of active layer is designed. The top electrode is then placed, and the encapsulating layer is a mixture of photosensitive crosslinking agent Cin–Cell and DtFDA.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

1. E. Bihar, Y. Deng, T. Miyake, M. Saadaoui, G. G. Malliaras, M. Rolandi, *Sci. Rep.* **2016**, *6*, 27582.
2. A. Koklu, S. Wustoni, V. E. Musteata, D. Ohayon, M. Moser, I. McCulloch, S. P. Nunes, S. Inal, *ACS Nano* **2021**, *15*, 8130.
3. H. Liu, A. Yang, J. Song, N. Wang, P. Lam, Y. Li, H. K. Law, F. Yan, *Sci. Adv.* **2021**, *7*, eabg8387.
4. J. Aerathupalathu Janardhanan, Y. L. Chen, C. T. Liu, H. S. Tseng, P. I. Wu, J. W. She, Y. S. Hsiao, H. H. Yu, *Anal. Chem.* **2022**, *94*, 7584.
5. M. Braendlein, T. Lonjaret, P. Leleux, J. M. Badier, G. G. Malliaras, *Adv. Sci.* **2017**, *4*, 1600247.
6. S. Park, S. W. Heo, W. Lee, D. Inoue, Z. Jiang, K. Yu, H. Jinno, D. Hashizume, M. Sekino, T. Yokota, K. Fukuda, K. Tajima, T. Someya, *Nature* **2018**, *561*, 516.
7. S. Han, S. Yu, S. Hu, H.-J. Chen, J. Wu, C. Liu, *J. Mater. Chem. C* **2021**, *9*, 11801.
8. S. K. Lee, Y. W. Cho, J. S. Lee, Y. R. Jung, S. H. Oh, J. Y. Sun, S. Kim, Y. C. Joo, *Adv. Sci.* **2021**, *8*, 2001544.
9. Y. Zhang, E. R. W. van Doremaele, G. Ye, T. Stevens, J. Song, R. C. Chiechi, Y. van de Burgt, *Adv. Mater.* **2022**, *34*, e2200393.
10. E. Zeglio, O. Inganas, *Adv. Mater.* **2018**, *30*, e1800941.
11. N. A. Kukhta, A. Marks, C. K. Luscombe, *Chem. Rev.* **2022**, *122*, 4325.
12. Y. Lei, P. Li, Y. Zheng, T. Lei, *Mater. Chem. Front.* **2024**, *8*, 133–158.
13. S. M. Yu, C. J. Kousseff, C. B. Nielsen, *Synth. Met.* **2023**, *293*, 117295.
14. E. Stein, O. Nahor, M. Stolov, V. Freger, I. M. Petruta, I. McCulloch, G. L. Frey, *Nat. Commun.* **2022**, *13*, 5548.
15. A. Giovannitti, C. B. Nielsen, D. T. Sbircea, S. Inal, M. Donahue, M. R. Niazi, D. A. Hanifi, A. Amassian, G. G. Malliaras, J. Rivnay, I. McCulloch, *Nat. Commun.* **2016**, *7*, 13066.
16. Y. Zhang, G. Ye, T. P. A. van der Pol, J. Dong, E. R. W. van Doremaele, I. Krauhausen, Y. Liu, P. Gkoupidenis, G. Portale, J. Song, R. C. Chiechi, Y. van de Burgt, *Adv. Funct. Mater.* **2022**, *32*, 2201593.
17. R. B. Rashid, W. Du, S. Griggs, I. P. Maria, I. McCulloch, J. Rivnay, *Sci. Adv.* **2021**, *7*, eabh1055.
18. J. J. Samuel, A. Garudapalli, A. A. Mohapatra, C. Gangadharappa, S. Patil, N. P. B. Aetukuri, *Adv. Funct. Mater.* **2021**, *31*, 2102903.
19. A. Wadsworth, H. Chen, K. J. Thorley, C. Cendra, M. Nikolka, H. Bristow, M. Moser, A. Salleo, T. D. Anthopoulos, H. Sirringhaus, I. McCulloch, *J. Am. Chem. Soc.* **2020**, *142*, 652.
20. Y. Li, M. Gu, Z. Pan, B. Zhang, X. Yang, J. Gu, Y. Chen, *J. Mater. Chem. A* **2017**, *5*, 10798.
21. Y. Liu, H. Phan, T. S. Herng, T. Y. Gopalakrishna, J. Ding, J. Wu, *Chem. Asian J.* **2017**, *12*, 2177.
22. X. X. Chen, J. T. Li, Y. H. Fang, X. Y. Deng, X. Q. Wang, G. Liu, Y. Wang, X. Gu, S. D. Jiang, T. Lei, *Nat. Commun.* **2022**, *13*, 2258.
23. J. Brodsky, I. Gablech, L. Migliaccio, M. Havlicek, M. J. Donahue, E. D. Glowacki, *ACS Appl. Mater.* **2023**, *15*, 27002.
24. W. Huang, J. Chen, Y. Yao, D. Zheng, X. Ji, L. W. Feng, D. Moore, N. R. Glavin, M. Xie, Y. Chen, R. M. Pankow, A. Surendran, Z. Wang, Y. Xia, L. Bai, J. Rivnay, J. Ping, X. Guo, Y. Cheng, T. J. Marks, A. Facchetti, *Nature* **2023**, *613*, 496.
25. S. Y. Jeong, J. W. Moon, S. Lee, Z. Wu, S. H. Park, J. H. Cho, H. Y. Woo, *Adv. Electron. Mater.* **2023**, *9*, 2300053.
26. D. U. Lim, S. B. Jo, J. H. Cho, *Adv. Mater.* **2023**, *35*, e2208757.
27. J. W. Moon, D. U. Lim, Y. N. Kim, J. H. Kim, J. H. Kim, S. B. Jo, J. H. Cho, *Adv. Funct. Mater.* **2023**, *33*, 2305440.
28. B. Wang, Y. Kong, S. Zhang, Z. Wu, S. Wang, J. Ren, H. Y. Woo, Y. Li, W. Ma, *Adv. Funct. Mater.* **2024**, *34*, 2312822.
29. J. Kim, R. M. Pankow, Y. Cho, I. D. Duplessis, F. Qin, D. Meli, R. Daso, D. Zheng, W. Huang, J. Rivnay, T. J. Marks, A. Facchetti, *Nat. Electron.* **2024**, *7*, 234.
30. M. Frisch, G. Trucks, H. Schlegel, G. Scuseria, M. Robb, J. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G. Petersson, *Gaussian*

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09 (Revision D.01), **2009**. https://www.scienceopen.com/document?vid=2f88089c-f1ad-492a-816b-4ce3ddc41530fGaussian 09 (Revision C.01) – ScienceOpen
31. K. Raghavachari, *Theor. Chem. Acc.* **2000**, *103*, 361.
32. S. T. M. Tan, S. Keene, A. Giovannitti, A. Melianas, M. Moser, I. McCulloch, A. Salleo, *J. Mater. Chem. C* **2021**, *9*, 12148.
33. Y. Q. Zheng, Y. Liu, D. Zhong, S. Nikzad, S. Liu, Z. Yu, D. Liu, H. C. Wu, C. Zhu, J. Li, H. Tran, J. B. Tok, Z. Bao, *Science* **2021**, *373*, 88.
34. Z. Wang, X. J. Zheng, T. Ouchi, T. B. Kouznetsova, H. K. Beech, S. Av-Ron, T. Matsuda, B. H. Bowser, S. Wang, J. A. Johnson, J. A. Kalow, B. D. Olsen, J. P. Gong, M. Rubinstein, S. L. Craig, *Science* **2021**, *374*, 193.
35. H. Chen, A. Wadsworth, C. Ma, A. Nanni, W. Zhang, M. Nikolka, A. M. T. Luci, L. M. A. Perdigao, K. J. Thorley, C. Cendra, B. Larson, G. Rumbles, T. D. Anthopoulos, A. Salleo, G. Costantini, H. Sirringhaus, I. McCulloch, *J. Am. Chem. Soc.* **2019**, *141*, 18806.
36. C. Liu, Y. Xu, Y.-Y. Noh, *Mater. Today* **2015**, *18*, 79.
37. Z. Zhou, X. Wu, T. L. D. Tam, C. G. Tang, S. Chen, K. Hou, T. Li, Q. He, J. J. Sit, J. Xu, W. L. Leong, *Adv. Funct. Mater.* **2023**, *34*, 2305780.
38. Z. Zhao, Z. Tian, F. Yan, *Cell Rep. Phys. Sci.* **2023**, *4*, 101673.
39. Y. Yao, W. Huang, J. Chen, G. Wang, H. Chen, X. Zhuang, Y. Ying, J. Ping, T. J. Marks, A. Facchetti, *Proc. Natl. Acad. Sci. U. S. A.* **2021**, *118*, e2111790118.
40. Y. Wang, G. Zhu, E. Zeglio, T. C. H. Castillo, S. Haseena, M. K. Ravva, S. Cong, J. Chen, L. Lan, Z. Li, A. Herland, I. McCulloch, S. Inal, W. Yue, *Chem. Mater.* **2023**, *35*, 405.

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